

Minor- II: Probability and Statistics

College For Integrated Studies
University of Hyderabad

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Duration: 60 minutes
Maximum Score: 20 points

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Course Code: MM 212

Instructions: No mark will be given for writing only TRUE or FALSE (without justification) in question 1. Answers without justification will receive a score of zero.

1. Suppose we toss two fair dice.

Let $E_1 :=$ the event that the sum of the dice is six,

$E_2 :=$ the event that sum of the dice equals seven,

and

$F :=$ the event that the first die equals four.

(a) True / False: E_1 independent of F .

[5]

(b) True / False: E_2 independent of F .

[5]

2. A type **C** battery is in working condition with probability $\frac{7}{10}$, whereas a type **D** battery is in working condition with probability $\frac{4}{10}$. A battery is randomly chosen from a bin consisting of 8 type **C** and 6 type **D** batteries. Given that the battery does not work, what is the probability that it was a type **C** battery.

[5]

3. Two gamblers **A** and **B** bet on the outcomes of successive flips of a **fair** coin. On each flip, if the coin comes up heads, **A** collects 1 unit from **B**, whereas if it comes up tails, **A** pays 1 unit to **B**. They continue to do this until one of them runs out of money. If it is assumed that the successive flips of the fair coin are independent. What is the probability that **A** ends up with all the money if he starts with k units and **B** starts with $N - k$ units ?

[5]

All the best !